

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

Keywords: Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

1 Introduction

1.1 Motivation and Context

Project portfolio management (PPM) is a fundamental organizational capability for maintaining competitive advantage and achieving strategic objectives. In project-based industries, portfolio decisions determine which initiatives are selected, prioritized, and funded

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under limited resources, conflicting criteria, and uncertain execution outcomes (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Martinsuo, 2013; Killen et al., 2008). These decisions become particularly critical in large-scale engineering environments, where organizations must coordinate multiple concurrent projects while balancing strategic value, financial feasibility, schedule pressure, and risk exposure.

In the Engineering, Procurement, and Construction (EPC) sector, especially in oil and gas projects, portfolio budgeting is complicated by heterogeneous contract structures, long project durations, capital intensity, and uncertain cash-flow behavior (Merrow, 2011; Cui et al., 2010; Kenley and Wilson, 1986; Kaka and Price, 1993). Budget allocation must account for both cash outflows associated with procurement, labor, equipment, and sub-contractors, and delayed cash inflows tied to contractual milestones and client approvals. Consequently, portfolio budgeting is not merely a one-time allocation problem; it is a sequential decision process in which current funding choices influence future project progress, liquidity conditions, and portfolio-level performance.

This challenge is amplified by persistent uncertainty in project execution. Large industrial and infrastructure projects frequently experience delays and cost overruns, and such deviations can propagate through both schedules and financial plans (Flyvbjerg et al., 2002, 2003; Love et al., 2013, 2016). Earned value indicators such as the Schedule Performance Index (SPI) and Cost Performance Index (CPI) provide a compact representation of execution performance and can support project monitoring and control (Fleming and Koppelman, 2016; Lipke et al., 2009; Vanhoucke, 2012). However, these indicators also underscore the difficulty of making reliable budget allocation decisions when future project states, payment realizations, and performance deviations remain uncertain. Therefore, portfolio budgeting under uncertain cash flows requires methods that can reason across multiple projects, multiple time periods, and stochastic state transitions.

The broader literature on project portfolio management similarly characterizes such environments by large data volumes, multiple decision criteria, resource constraints, and the need for investment optimization. These characteristics are consistent with prior research on project portfolio selection, project scheduling under uncertainty, and construction cash-flow forecasting (Gutjahr and Froeschl, 2013; Doerner et al., 2006; Khanzadi et al., 2018; Golenko-Ginzburg and Gonik, 1997; Ballestín and Blanco, 2015; Barraza et al., 2007). Taken together, they motivate the development of intelligent decision-support methods when static or manually driven budgeting approaches become insufficient.

1.2 Research Gap

A substantial body of literature has addressed project portfolio selection, prioritization, and scheduling through operations research, multi-criteria decision-making, stochastic programming, and metaheuristic optimization (Archer and Ghasemzadeh, 1999; Liesiö et al., 2007; Klerides and Hadjiconstantinou, 2019; Elshaer, 2014). These methods provide valuable structure for complex decision problems, but they are often less suitable when portfolio budget allocation must respond to evolving project conditions under uncertainty. In many existing formulations, project cash flows, budget requirements, and performance trajectories are treated as fixed inputs, deterministic plans, or exogenous scenarios. This limits their ability to capture the interaction between current allocation decisions and future project and financial states.

A central limitation in this context is that portfolio budgeting in EPC environments is inherently sequential. Budget releases made at one time step affect project execution, milestone realization, cash availability, and subsequent allocation opportunities. Yet much of the existing literature approaches project portfolio decisions as static optimization,

stage-wise planning, or scheduling problems rather than as a learned sequential policy problem under uncertainty. As a result, the adaptive linkage between observed portfolio state and subsequent allocation action remains insufficiently modeled.

Recent advances in machine learning and reinforcement learning provide a promising direction for this class of problems. Reinforcement learning is specifically designed for sequential decision-making under uncertainty and has demonstrated value in operations, scheduling, and process control settings (Puterman, 1994; Sutton and Barto, 2018; Wauters et al., 2016; Chen et al., 2018; Hubbs et al., 2020; Bengio et al., 2021). Nevertheless, its application to project portfolio budgeting under uncertain cash flows remains limited. There is still a need for a reproducible framework that can integrate project-level uncertainty, cash-flow timing, milestone-based payments, and adaptive resource allocation within a portfolio-level sequential decision architecture.

A further practical constraint is data availability. Detailed industrial project portfolio data are typically confidential and difficult to release publicly, which creates a barrier to reproducible experimentation. As a result, the literature lacks sufficiently transparent benchmark environments for testing adaptive portfolio budgeting policies. This gap motivates the use of literature-calibrated synthetic data as a practical foundation for developing and evaluating learning-based decision-support systems (Jordon et al., 2022; Xu et al., 2019).

1.3 Proposed Approach and Contributions

To address these gaps, this paper develops a deep reinforcement learning framework for optimal project portfolio budgeting under uncertain cash flows. The problem is modeled as a sequential decision process in which an agent observes the evolving state of a project portfolio and allocates constrained financial resources over time. The state representation incorporates project-level and portfolio-level information, including budget at completion, planned duration, contractual cash-flow patterns, performance indicators, milestone-based payment structures, and liquidity-related variables.

The framework is designed to learn an allocation policy directly from repeated interaction with a stochastic simulation environment. Rather than relying on predefined deterministic decision rules or repeated re-optimization procedures, the agent learns how present allocation choices affect future portfolio states and cumulative performance over time. To learn this policy, the study employs deep reinforcement learning, with Proximal Policy Optimization (PPO) as the primary training algorithm (Schulman et al., 2017), consistent with the broader reinforcement-learning literature on stochastic control and policy optimization (Sutton and Barto, 2018; Haarnoja et al., 2018).

After training, the learned policy can also be deployed within the simulation environment to generate a forward policy rollout over a specified horizon, such as the next 12 time steps, in order to illustrate plausible portfolio evolution and corresponding recommended allocations. This forward policy rollout should be interpreted as a simulated policy-driven trajectory rather than as a realized future observation. Its purpose is to demonstrate the behavioral implications of the learned allocation policy under the modeled environment dynamics until actual future project states become available.

Because real EPC portfolio datasets are typically inaccessible due to confidentiality restrictions, the framework uses a literature-calibrated synthetic data generation process. This allows reproducible experimentation while preserving consistency with known characteristics of project cash flows, schedule variability, and portfolio structure. The synthetic environment is intended as a research testbed that can later be refined and calibrated using anonymized organizational data.

The main contributions of this paper are as follows:

1. **A sequential formulation of project portfolio budgeting:** The study formulates EPC portfolio budgeting under uncertain cash flows as a dynamic decision problem in which allocations, project progress, and liquidity conditions evolve jointly over time.
2. **A custom reinforcement learning environment for portfolio budgeting:** The paper develops an environment that captures project-level uncertainty, cash inflows and outflows, milestone-based payments, and portfolio-level financial constraints.
3. **A deep reinforcement learning policy for adaptive allocation:** The framework trains an agent to learn budget allocation decisions directly from stochastic interaction with the environment.
4. **A forward policy rollout capability for trajectory demonstration:** The study shows how the trained policy can be used to simulate plausible future portfolio trajectories and recommended allocations over a finite horizon.
5. **A literature-calibrated synthetic data framework:** The study provides a reproducible synthetic portfolio-generation procedure to address the industrial data confidentiality barrier.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews prior research on project portfolio management, project control under uncertainty, and reinforcement learning for sequential decision-making. Section 3 formulates the project portfolio budgeting problem under uncertain cash flows as a sequential stochastic decision problem. Section 5 presents the proposed solution methodology, including the synthetic data generation framework, cash-flow and uncertainty modeling, reinforcement learning environment design, and PPO-based training procedure. Section 6 reports the computational results and analysis, including benchmark comparisons, robustness assessment, and forward policy rollout analysis. Finally, Section 7 concludes the paper and outlines future research directions.

2 Literature Review

2.1 Project Portfolio Management Under Uncertainty

Project portfolio management (PPM) has been extensively studied in operations research and management science. Early work by Archer and Ghasemzadeh (1999) established frameworks for project selection and prioritization, while Cooper et al. (2001) developed stage-gate processes for new product development portfolios. However, these approaches primarily address the *selection* problem (which projects to pursue) rather than the *execution* problem (how to allocate resources dynamically).

? formulated multi-objective portfolio optimization models balancing expected return, risk, and strategic alignment. Doerner et al. (2006) applied Pareto ant colony optimization to generate efficient frontiers for portfolio selection. These methods assume deterministic project execution once selected, ignoring the sequential decision-making required under performance uncertainty.

Stochastic programming approaches have been proposed to handle uncertainty. [Golenko-Ginzburg and Gonik \(1997\)](#) modeled resource-constrained project scheduling with stochastic activity durations. [Klerides and Hadjiconstantinou \(2019\)](#) developed two-stage stochastic programs for portfolio selection under demand uncertainty. However, these methods require explicit scenario trees and known probability distributions, which are unavailable for contractor performance dynamics.

2.2 Project Performance Uncertainty and Empirical Evidence

A substantial body of empirical research documents systematic cost and schedule overruns in capital projects. [Flyvbjerg et al. \(2002\)](#) analyzed 258 transportation infrastructure projects, finding mean cost overruns of 28% and schedule delays of 27%. [Flyvbjerg et al. \(2003\)](#) extended this analysis to 258 projects across multiple sectors, confirming left-skewed performance distributions with long tails toward poor outcomes.

[Merrow \(2011\)](#) provided the most comprehensive analysis of EPC project performance, examining 318 industrial megaprojects in oil & gas, chemicals, and mining. Key findings include: (1) mean SPI at completion of 0.81 (19% behind schedule), (2) mean CPI of 0.87 (15% over budget), (3) positive correlation between schedule and cost performance ($\rho \approx 0.68$), and (4) systematic variation in performance by project size, complexity, and contractor experience.

[Love et al. \(2013\)](#) fitted probability distributions to schedule overrun data from 276 construction projects, validating Beta and lognormal distributions as appropriate models. [Love et al. \(2016\)](#) analyzed the relationship between SPI and CPI, confirming that schedule delays typically precede cost overruns due to extended overhead and acceleration costs.

2.3 Cashflow Modeling and S-Curves

Project cashflow modeling has a rich history in construction management. [Kenley and Wilson \(1986\)](#) introduced the idiographic approach to cashflow forecasting, recognizing that each project exhibits unique spending patterns. [Kaka and Price \(1993\)](#) developed neural network models for cashflow prediction based on historical project data.

The S-curve model, representing cumulative spending as a sigmoid function of project duration, has become the standard representation. [Cioffi \(2005\)](#) provided analytical parameterization of S-curves using Beta distributions, enabling closed-form expressions for cashflow velocity and peak spending timing. [Barraza et al. \(2007\)](#) extended this to probabilistic S-curves incorporating uncertainty in project duration and spending patterns.

[Cui et al. \(2010\)](#) analyzed milestone-based payment structures in construction contracts, finding that 87% of contracts use progress-based payments rather than time-based schedules. This finding motivates our modeling choice to link payment timing to actual progress (SPI) rather than calendar dates.

2.4 Reinforcement Learning in Operations Research

RL has emerged as a powerful tool for sequential decision-making under uncertainty. [Sutton and Barto \(2018\)](#) provide the foundational textbook covering Markov Decision Processes (MDPs), temporal difference learning, and policy gradient methods. Recent advances in deep RL, particularly Proximal Policy Optimization (PPO) ([Schulman et al., 2017](#)) and Soft Actor-Critic (SAC) ([Haarnoja et al., 2018](#)), have enabled application to high-dimensional continuous control problems.

Applications of RL to operations research problems have grown rapidly. [Hubbs et al. \(2020\)](#) applied deep RL to chemical production scheduling, achieving near-optimal solutions with 1000x speedup over MIP. [Bengio et al. \(2021\)](#) surveyed machine learning for combinatorial optimization, highlighting RL’s ability to learn heuristics that generalize across problem instances.

In project management, RL applications remain limited. [Wauters et al. \(2016\)](#) used RL for resource-constrained project scheduling with deterministic durations. [Chen et al. \(2018\)](#) applied RL to construction project scheduling under resource uncertainty. However, no prior work addresses portfolio-level budgeting under cashflow uncertainty with stochastic contractor performance.

2.5 Rolling Horizon Control

Rolling horizon control (also called receding horizon control or model predictive control) is a well-established technique in control theory and operations research. [Sethi and Thompson \(1991\)](#) surveyed rolling horizon procedures for production planning, demonstrating their effectiveness in handling long planning horizons with bounded computational effort.

[Chand et al. \(2002\)](#) analyzed the theoretical properties of rolling horizon heuristics, proving asymptotic optimality under certain conditions. [Sahin et al. \(2013\)](#) applied rolling horizon frameworks to inventory management under demand uncertainty, showing that fixed-horizon re-planning can approximate infinite-horizon optimal policies.

In project management, [Lambrechts et al. \(2008\)](#) used rolling horizon scheduling for resource-constrained projects under activity duration uncertainty. [Ballestín and Blanco \(2015\)](#) applied rolling horizon methods to multi-project resource allocation. However, these approaches use deterministic optimization at each re-planning step, rather than learning adaptive policies through RL.

2.6 Synthetic Data Generation for OR/ML Research

The use of synthetic data in operations research has a long tradition. [Kolisch and Sprecher \(1997\)](#) created the Project Scheduling Problem Library (PSPLIB), a benchmark dataset of synthetic instances that became the standard for algorithm comparison. [?](#) generated synthetic project networks to study schedule risk analysis methods.

In machine learning, synthetic data generation has become essential for domains with data scarcity or privacy constraints. [Jordon et al. \(2022\)](#) surveyed synthetic data generation methods, emphasizing the importance of preserving statistical properties of real data. [Xu et al. \(2019\)](#) developed generative models for tabular data that maintain correlations and distributional characteristics.

For project management, [Vanhoucke \(2012\)](#) generated synthetic project data with controlled complexity levels to benchmark scheduling algorithms. [Elshaer \(2014\)](#) created synthetic portfolios with varying risk profiles to test portfolio optimization methods. Our work extends this tradition by developing a comprehensive synthetic data generator calibrated to empirical distributions from multiple literature sources, ensuring realism while maintaining reproducibility.

2.7 Research Gaps

Despite extensive research in project portfolio management, performance uncertainty, and RL applications, critical gaps remain:

1. **No RL framework for portfolio budgeting:** Existing RL applications to project management focus on single-project scheduling, not portfolio-level resource allocation under cashflow uncertainty.
2. **Lack of realistic synthetic environments:** Prior synthetic data generators for project management do not model stochastic contractor performance, milestone-based payments, or portfolio-level cashflow dynamics.
3. **Absence of rolling horizon RL architectures:** Rolling horizon control has not been integrated with RL for portfolio management, despite its natural fit for handling variable-duration portfolios.
4. **No methodology for literature-calibrated data generation:** Existing approaches either use proprietary data (unavailable for research) or arbitrary synthetic distributions (lacking empirical grounding).

This paper addresses these gaps by developing the first RL framework for portfolio budgeting with a literature-calibrated synthetic environment and rolling horizon architecture.

3 Problem Formulation

4 Problem Formulation

4.1 Management Problem and Portfolio Definition

We consider a multi-period budget allocation and operational control problem for a fixed portfolio of contracted projects. The portfolio is defined as the aggregation of a finite set of projects,

$$P = \{1, 2, \dots, n\},$$

where each project $i \in P$ represents a contractual undertaking with its own budget, planned schedule, expenditure profile, payment structure, and performance state. The portfolio is not a decision variable in the sense of project selection. Instead, it is the set of ongoing and committed projects whose execution must be managed jointly under limited financial and operational resources.

The planning horizon is indexed by discrete periods $t = 1, \dots, H$. At each period, management observes the current portfolio condition, allocates financial resources across active projects, and may apply corrective interventions to improve schedule or cost performance. Projects consume resources over time, generate cash inflows through milestone-based payments, and may deviate from their contractual plans because of schedule delays, cost overruns, funding shortages, payment delays, contractor performance uncertainty, or operational disruptions.

The central management problem is to determine a sequential policy for budget allocation and intervention decisions that balances multiple objectives: completing projects close to their planned finish dates, preserving profitability, avoiding excessive borrowing, reducing delay penalties, limiting wasteful expenditure on economically unattractive projects, and maintaining acceptable portfolio-level financial health. Different organizations may prioritize these goals differently. For example, some may prefer on-time delivery even at high cost, while others may prioritize profitability and reduce funding to projects whose expected continuation value becomes weak.

A key feature of the problem is that budget allocation and intervention decisions do not produce known deterministic performance outcomes. After management allocates resources and applies an intervention decision in period t , the contractor's realized response is uncertain and is only observed at the next reporting epoch through project-control information such as realized progress, earned value, schedule performance, cost performance, and updated remaining-duration estimates. This makes the problem naturally sequential and feedback-driven.

4.2 Contractual Project Attributes

Each project $i \in P$ is first described by contractual and planned attributes known at the beginning of the planning process. Let s_i and f_i denote the planned start and planned finish periods, respectively. The planned duration is

$$D_i^{\text{plan}} = f_i - s_i + 1,$$

and the contractual planned execution window is

$$\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}.$$

The budget at completion of project i is denoted by BAC_i . This value represents the planned cost basis of the project and is used to define planned expenditure, contract revenue, intervention cost scaling, and credit exposure.

The contract revenue of project i is

$$CR_i = \text{BAC}_i(1 + \pi_i),$$

where π_i is the contractual profit margin. The notation CR_i is used for contract revenue to avoid conflict with remaining-duration variables.

4.3 Planned Expenditure and Contractual S-Curve

The planned cumulative expenditure of each project follows a normalized S-curve. Since the model is period-based, planned cumulative quantities are interpreted at the end of each period. The normalized planned time is therefore defined as

$$\tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad t \in \mathcal{A}_i^{\text{plan}}.$$

Thus, $\tau_{i,s_i} = 1/D_i^{\text{plan}}$ and $\tau_{i,f_i} = 1$. For notational completeness, $\tau_{i,t}$ is truncated to the unit interval when needed:

$$\tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}.$$

The planned cumulative expenditure fraction is modeled as

$$F_i(\tau) = I_\tau(a_i, b_i),$$

where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function and $a_i, b_i > 0$ are project-specific shape parameters. The planned cumulative expenditure is

$$\bar{E}_{i,t} = \text{BAC}_i F_i(\tau_{i,t}^c),$$

and the planned period expenditure is

$$\bar{e}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1}.$$

This S-curve represents the contractual baseline spending trajectory. It is used as a benchmark for measuring planned progress, schedule deviation, and performance indices. It does not force actual execution to stop at the planned finish date. If the planned progress has not been achieved, the unfinished work is carried forward to subsequent periods. Consequently, actual project completion time and portfolio completion time are endogenous outcomes determined by realized project progress.

4.4 Actual Project State and Performance History

Each project has an actual execution state that may deviate from its contractual baseline. The realized active project set at time t is defined by actual completion status:

$$P_t = \{i \in P : s_i \leq t, p_{i,t} < 1\},$$

where $p_{i,t} \in [0, 1]$ is the realized physical progress of project i by the end of period t . This definition allows a project to remain active after its planned finish date if the realized progress has not reached completion.

A project is completed only when realized progress reaches the completion threshold:

$$p_{i,t} \geq 1.$$

The actual completion period is therefore

$$T_i^{\text{comp}} = \min\{t : p_{i,t} \geq 1\}.$$

The portfolio completion period is

$$T^{P,\text{comp}} = \max_{i \in P} T_i^{\text{comp}}.$$

If project i is not completed within the modeled horizon H , its completion time may be treated as censored. In that case, terminal delay, residual value, and remaining exposure may be evaluated using the latest remaining-duration estimate $RD_{i,H}$, an empirical terminal-value approximation, or the terminal convention adopted in the implementation.

The planned finish period f_i is therefore a contractual benchmark, not a stopping rule. If

$$t > f_i \quad \text{and} \quad p_{i,t} < 1,$$

then project i is delayed and remains active.

The planned cumulative progress benchmark is denoted by

$$\bar{p}_{i,t} = F_i(\tau_{i,t}^c).$$

The progress gap is

$$G_{i,t}^p = \max\{0, \bar{p}_{i,t} - p_{i,t}\}.$$

Thus, $G_{i,t}^p > 0$ indicates that the project is behind its planned progress curve.

The earned value of project i at time t may be represented as

$$EV_{i,t} = \text{BAC}_i p_{i,t},$$

or, more generally, by the earned-value measure reported by the organization's project-control system.

The schedule performance index may be defined as

$$\text{SPI}_{i,t} = \frac{p_{i,t}}{\max\{\epsilon, \bar{p}_{i,t}\}},$$

where $\epsilon > 0$ avoids division by zero in early periods. Values below one indicate schedule underperformance relative to the contractual S-curve.

Similarly, the cost performance index $\text{CPI}_{i,t}$ measures cost efficiency relative to earned progress. If earned-value accounting data are available, one possible definition is

$$\text{CPI}_{i,t} = \frac{\text{EV}_{i,t}}{\max\{\epsilon, \text{AC}_{i,t}\}},$$

where $\text{AC}_{i,t}$ is the actual cost accumulated up to period t . The precise empirical definition of CPI may depend on the available project accounting system. In general, it summarizes whether the realized cost of earned work is above or below the planned cost basis.

The condition of project i at time t is represented by

$$\Omega_{i,t} = (p_{i,t}, \text{EV}_{i,t}, \text{SPI}_{i,t}, \text{CPI}_{i,t}, \text{RD}_{i,t}, h_{i,t}, z_{i,t}^{\text{rep}}),$$

where the variables are defined in [Table 1](#).

Table 1: Project state and performance variables

Symbol	Description	Range
$p_{i,t}$	Realized physical progress	$[0, 1]$
$\text{EV}_{i,t}$	Earned value or reported earned-work measure	$\mathbb{R}_{\geq 0}$
$\text{SPI}_{i,t}$	Schedule performance index	$\mathbb{R}_{> 0}$
$\text{CPI}_{i,t}$	Cost performance index	$\mathbb{R}_{> 0}$
$\text{RD}_{i,t}$	Reported or estimated remaining duration based on project-control data	$\mathbb{Z}_{\geq 0}$
$h_{i,t}$	Intervention-persistence state	discrete or continuous
$z_{i,t}^{\text{rep}}$	Replanning indicator	$\{0, 1\}$

The variable $\text{RD}_{i,t}$ is not assumed to be generated by a deterministic duration-transition model. It represents a reported, forecasted, or management-estimated remaining duration based on the latest project-control information available at decision epoch t .

The expenditure allocated to project i in period t is denoted by $e_{i,t}$. This expenditure may differ from the planned value $\bar{e}_{i,t}$ because of resource constraints, inefficiencies, acceleration, rework, disruption, or management intervention. The allocation $e_{i,t}$ is a management action. Its effect on future project performance is uncertain and is mediated by the contractor’s realized response, which becomes observable only in subsequent project reports.

The model does not impose a deterministic structural transition law for actual progress. Instead, realized progress $p_{i,t}$ is treated as an observed execution state at each decision epoch. This distinction is important: the contractual S-curve defines the planned benchmark, while actual project progress is determined by real execution and reported through project-control data. The management model uses the observed state to allocate resources and evaluate performance; it does not require an artificial law forcing the project to move along the baseline schedule.

Therefore, if actual progress falls short of planned progress, the project is not analytically “transitioned” to completion. It simply remains incomplete, continues to appear in the active set P_t , and may generate delay exposure, financing needs, intervention costs, and future cash-flow consequences.

4.5 Milestone-Based Revenue

Each project receives revenue through a milestone-based payment plan. Let

$$K_i = \{0, 1, \dots, N_i\}$$

denote the set of payment milestones for project i , where $k = 0$ may represent an advance payment and $k = N_i$ represents the final payment.

Each milestone has a payment fraction $q_{i,k} \geq 0$, satisfying

$$\sum_{k \in K_i} q_{i,k} = 1.$$

The gross payment associated with milestone k is

$$P_{i,k}^{\text{gross}} = q_{i,k} C R_i.$$

Milestone payments are triggered by realized progress. Let $\chi_{i,k} \in [0, 1]$ denote the progress threshold required for milestone k . Under the period-end S-curve convention, the planned milestone achievement time is

$$t_{i,k}^{\text{plan}} = s_i + \chi_{i,k} D_i^{\text{plan}} - 1.$$

This follows from the normalized period-end relation

$$\tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}},$$

so that the planned period corresponding to progress threshold $\chi_{i,k}$ satisfies

$$\chi_{i,k} = \frac{t_{i,k}^{\text{plan}} - s_i + 1}{D_i^{\text{plan}}}.$$

The value $t_{i,k}^{\text{plan}}$ may be fractional and is interpreted as a continuous planned benchmark time. When a discrete planned period is required, the mapping $\mathcal{T}(\cdot)$ is used.

The actual achievement time is the first period in which realized progress reaches the required threshold:

$$t_{i,k}^{\text{act}} = \min\{t : p_{i,t} \geq \chi_{i,k}\}.$$

A client-side payment processing delay $\delta_{i,k} \geq 0$ shifts milestone achievement into cash receipt time:

$$t_{i,k}^{\text{rec}} = t_{i,k}^{\text{act}} + \delta_{i,k}.$$

For advance payments, the receipt occurs at project start and no delay is imposed:

$$t_{i,0}^{\text{rec}} = s_i.$$

Because the model operates in discrete planning periods, continuous receipt times are mapped to accounting periods using

$$\mathcal{T}(x) = \lceil x \rceil.$$

Thus, if a milestone is achieved between two periods, the receipt is recognized in the later period.

4.6 Retention Mechanism

Retention is modeled as a contractual withholding mechanism applied to interim progress payments. Let $r_i \in [0, 1]$ denote the retention rate for project i . The advance payment, if present, is treated separately from retention withholding.

For the advance milestone $k = 0$, the net receipt is

$$P_{i,0}^{\text{net}} = P_{i,0}^{\text{gross}}.$$

For interim non-final progress milestones $k = 1, \dots, N_i - 1$, the net receipt is

$$P_{i,k}^{\text{net}} = (1 - r_i)P_{i,k}^{\text{gross}}.$$

The accumulated retention prior to final completion is

$$R_i^{\text{ret}} = \sum_{j=1}^{N_i-1} r_i P_{i,j}^{\text{gross}}.$$

The final milestone is not subject to additional retention withholding. At project completion, the final milestone payment is received together with the release of all accumulated retention:

$$P_{i,N_i}^{\text{net}} = P_{i,N_i}^{\text{gross}} + R_i^{\text{ret}}.$$

Therefore, total lifecycle cash receipts equal the contract revenue:

$$\sum_{k=0}^{N_i} P_{i,k}^{\text{net}} = C R_i.$$

The period revenue inflow from project i is

$$Y_{i,t} = \sum_{k \in K_i} P_{i,k}^{\text{net}} \mathbf{1} \{ \mathcal{T}(t_{i,k}^{\text{rec}}) = t \}.$$

The aggregate portfolio revenue inflow is

$$Y_t^P = \sum_{i \in P} Y_{i,t}.$$

A separate defects liability period is not modeled. Retention is assumed to be released at project completion because the planning horizon focuses on active project execution rather than post-completion warranty obligations.

4.7 Funding, Working Capital, and Credit Mechanism

At each period, the portfolio has exogenous available funding B_t^0 , revenue inflows Y_t^P , and access to external financing or credit. The available internal funding before additional borrowing is

$$B_t^{\text{avail}} = B_t^0 + Y_t^P.$$

The total required cash outflow is

$$O_t = \sum_{i \in P_t} e_{i,t} + C_t^{\text{act}}.$$

If required expenditure exceeds internally available funding, the excess is financed through credit rather than treated as a hard infeasibility. The credit drawdown is

$$L_t = \max\{0, O_t - B_t^{\text{avail}}\}.$$

The total funded budget is therefore

$$B_t = B_t^{\text{avail}} + L_t.$$

This structure removes the need for a hard funding constraint of the form

$$O_t \leq B_t^{\text{avail}}.$$

Budget shortfalls remain feasible, but they generate financing costs, liquidity risk, and possible credit-limit violations.

Project-level working capital is tracked through pre-penalty and post-penalty balances. The pre-penalty working capital position is

$$\widetilde{W}_{i,t} = W_{i,t-1} + Y_{i,t} - e_{i,t}.$$

Each project has a credit limit

$$CL_i = \theta_i^{\text{CL}} \text{BAC}_i (1 - \gamma_i A_i),$$

where θ_i^{CL} is a credit-limit factor, $A_i = q_{i,0}$ is the advance-payment fraction, and γ_i controls the extent to which advance payments reduce remaining credit capacity.

A credit-limit violation occurs when the negative working-capital position exceeds the available limit:

$$V_{i,t}^{\text{credit}} = \max\{0, -\widetilde{W}_{i,t} - CL_i\}.$$

The associated credit cost is

$$c_{i,t}^{\text{credit}} = \psi_i \left(V_{i,t}^{\text{credit}} \right),$$

where $\psi_i(\cdot)$ is a nonnegative increasing penalty function. The post-penalty working capital balance is

$$W_{i,t} = \widetilde{W}_{i,t} - c_{i,t}^{\text{credit}}.$$

The portfolio credit-violation cost is

$$C_t^{\text{credit}} = \sum_{i \in P_t} c_{i,t}^{\text{credit}}.$$

Borrowed funds may also incur financing cost. Let

$$C_t^{\text{fin}} = \varphi_t(L_t),$$

where $\varphi_t(\cdot)$ is a nonnegative increasing function representing interest, financing charges, or liquidity risk associated with external funding.

4.8 Delay Penalties and Contractual Consequences

Schedule delay is measured relative to contractual planned completion. The delay of project i at completion is

$$\Delta_i^{\text{delay}} = \max\{0, T_i^{\text{comp}} - f_i\}.$$

A delay penalty or liquidated damage cost is defined as the terminal project-level cost

$$C_i^{\text{LD}} = \ell_i \Delta_i^{\text{delay}},$$

where $\ell_i \geq 0$ is the contractual delay penalty rate.

In addition to terminal liquidated damages, ongoing schedule exposure may be measured during execution as

$$D_t = \sum_{i \in P_t} \max\{0, t - f_i\} \mathbf{1}\{p_{i,t} < 1\}.$$

This term captures the extent of active lateness among projects that remain incomplete after their planned finish dates. It is a period-level exposure measure, whereas C_i^{LD} is a terminal cost incurred after actual completion.

4.9 Intervention Management

Management may apply corrective interventions to improve project performance. Let

$$u_{i,t} \in \{0, 1\}$$

indicate whether an intervention is applied to project i in period t .

The schedule underperformance gap is

$$G_{i,t}^{\text{SPI}} = \max\{0, 1 - \text{SPI}_{i,t}\}.$$

The potential cost of activating an intervention is

$$\bar{C}_{i,t}^{\text{act}} = \kappa_i \text{BAC}_i G_{i,t}^{\text{SPI}},$$

where $\kappa_i \geq 0$ controls the intervention cost intensity. The realized intervention cost is incurred only if the intervention is activated:

$$C_{i,t}^{\text{act}} = u_{i,t} \bar{C}_{i,t}^{\text{act}}.$$

The total portfolio intervention cost is

$$C_t^{\text{act}} = \sum_{i \in P_t} C_{i,t}^{\text{act}}.$$

When an intervention is applied, it may improve the effective performance indicators used by management to evaluate the project condition. The effective schedule performance is represented as

$$\text{SPI}_{i,t}^{\text{eff}} = \text{SPI}_{i,t} + u_{i,t} \eta_{i,t}^{\text{SPI}} G_{i,t}^{\text{SPI}},$$

where $\eta_{i,t}^{\text{SPI}} \in [0, 1]$ is the schedule-improvement effectiveness. This specification improves delayed projects while avoiding artificial reductions in performance when $\text{SPI}_{i,t} > 1$.

Similarly, the effective cost performance may be represented as

$$\text{CPI}_{i,t}^{\text{eff}} = \text{CPI}_{i,t} (1 + u_{i,t} \eta_{i,t}^{\text{CPI}}),$$

where $\eta_{i,t}^{\text{CPI}} \geq 0$ captures potential cost-efficiency improvement.

The realized effect of an intervention is uncertain. The values $\eta_{i,t}^{\text{SPI}}$ and $\eta_{i,t}^{\text{CPI}}$ are therefore not assumed to be known deterministic constants at the time of decision. They represent uncertain intervention effectiveness, which may be estimated from historical data, calibrated in simulation, sampled from scenarios, or learned through interaction with the project-control environment.

Intervention effects may persist beyond the activation period and decay over time. This persistence can be summarized through the state variable $h_{i,t}$, which records the remaining influence of past interventions. The formulation does not require a fixed deterministic persistence law. Instead, $h_{i,t}$ is treated as part of the observed or estimated project-control state and may be updated through the implementation-specific state-generation mechanism.

The expected benefits of intervention may include avoided delay penalties, avoided cost overruns, and opportunity gains. These benefits are not necessarily added as separate reward terms unless management assigns direct value to them. In the base formulation, intervention benefits are realized indirectly through improved project condition, reduced delay exposure, lower financing needs, lower credit stress, and higher realized portfolio value. If desired, diagnostic intervention benefits can be computed as

$$B_{i,t}^{\text{LD}} = \ell_i \mathbb{E} [\Delta D_{i,t}^{\text{avoided}}], \quad (1)$$

$$B_{i,t}^{\text{cost}} = \text{BAC}_i \mathbb{E} \left[\frac{1}{\text{CPI}_{i,t}} - \frac{1}{\text{CPI}_{i,t}^{\text{eff}}} \right], \quad (2)$$

$$B_{i,t}^{\text{opp}} = \nu_i V_i \mathbb{E} [\Delta D_{i,t}^{\text{avoided}}], \quad (3)$$

where $\Delta D_{i,t}^{\text{avoided}}$ is the expected reduction in delay, V_i is a project value-rate measure, and ν_i is an opportunity-value coefficient. These diagnostic quantities can support managerial interpretation but are not required as separate objective terms in the base model.

4.10 Duration and Performance Uncertainty

Project duration, cost efficiency, schedule performance, payment timing, contractor productivity, and intervention effectiveness are uncertain. The formulation does not require a specific probability distribution for these uncertainties. Instead, realized completion emerges from observed progress:

$$T_i^{\text{comp}} = \min\{t : p_{i,t} \geq 1\}.$$

Performance variables such as $p_{i,t}$, $\text{EV}_{i,t}$, $\text{SPI}_{i,t}$, $\text{CPI}_{i,t}$, $\text{RD}_{i,t}$, and $h_{i,t}$ are updated from project-control information, funding history, intervention history, and realized shocks. The key operational timing is as follows: management selects budget allocation and intervention decisions during period t , the contractor responds through actual execution during that period, and the realized performance consequences are observed at the next reporting epoch.

Thus, given the current state and action, the next state may be conceptually represented as

$$\Omega_{i,t+1} \sim \mathcal{M}_i(\cdot \mid \Omega_{i,t}, e_{i,t}, u_{i,t}),$$

or, at the portfolio level,

$$\Omega_{t+1} \sim \mathcal{M}(\cdot \mid \Omega_t, a_t),$$

where $\mathcal{M}$ denotes the real project-control environment, empirical state-generation process, simulator, or scenario mechanism. Importantly, $\mathcal{M}$ is not assumed to be known in closed form at the formulation stage.

This distinction is central to the modeling approach. Traditional deterministic or distribution-explicit optimization models often require the decision maker to specify future transition equations or probability distributions in advance. In contrast, the present formulation only specifies the sequential information and decision structure. The uncertainty model can be calibrated later in the methodology using historical data, expert scenarios, Monte Carlo simulation, or learned from interaction in a reinforcement-learning environment.

The important modeling assumption is not that a specific formula physically determines future progress. Rather, the model assumes that actual progress and performance indicators are observed over time and used by management to make sequential resource-allocation decisions. Allocation and intervention decisions influence the uncertain contractor response, but the realized effect is revealed only after the subsequent progress report.

4.11 Sequential Decision Problem

At each period t , management observes the portfolio state

$$\Omega_t = \{\Omega_{i,t} : i \in P_t\},$$

the available internal funding B_t^{avail} , revenue inflows Y_t^P , credit exposure, delay exposure, and project performance history. The decision vector is

$$a_t = (\{e_{i,t}\}_{i \in P_t}, \{u_{i,t}\}_{i \in P_t}),$$

where $e_{i,t}$ is the expenditure allocated to project i , and $u_{i,t}$ is the intervention decision.

The basic feasibility bounds are

$$e_{i,t} \geq 0, \quad u_{i,t} \in \{0, 1\}, \quad 0 \leq p_{i,t} \leq 1.$$

Additional operational bounds may be imposed if needed, such as maximum period expenditure, intervention-capacity limits, or project-specific resource constraints:

$$0 \leq e_{i,t} \leq \bar{e}_{i,t}^{\max}, \quad \sum_{i \in P_t} u_{i,t} \leq \bar{U}_t^{\max}.$$

Unlike a hard-budget formulation, this model does not require

$$\sum_{i \in P_t} e_{i,t} + C_t^{\text{act}} \leq B_t^{\text{avail}}.$$

Instead, expenditures beyond internally available funds are financed through credit:

$$L_t = \max \left\{ 0, \sum_{i \in P_t} e_{i,t} + C_t^{\text{act}} - B_t^{\text{avail}} \right\}.$$

Thus, budget shortfalls remain feasible but costly. This structure reflects a contractor that can borrow, draw credit, or absorb liquidity risk, but must account for the financial consequences of doing so.

The decision process is non-anticipative: decisions at time t may depend only on information available up to and including period t . After the action a_t is implemented, the resulting contractor performance is not known immediately. The next-period state is revealed after one reporting interval through updated progress and performance reports:

$$\boldsymbol{\Omega}_{t+1} = \left\{ (p_{i,t+1}, \text{EV}_{i,t+1}, \text{SPI}_{i,t+1}, \text{CPI}_{i,t+1}, \text{RD}_{i,t+1}, h_{i,t+1}, z_{i,t+1}^{\text{rep}}) : i \in P_{t+1} \right\}.$$

Therefore, the action a_t affects the future state through an uncertain contractor response, but the realized outcome is observed only ex post. This is the core feedback structure of the sequential control problem.

The goal is to find a sequential policy

$$\pi = \{a_t\}_{t=1}^H$$

that balances schedule, cost, liquidity, profitability, and risk outcomes over the portfolio horizon. More generally, the policy can be written as a state-feedback mapping,

$$a_t = \pi(\boldsymbol{\Omega}_t, B_t^{\text{avail}}, Y_t^P, \dots),$$

where decisions are adapted to the information available at the current decision epoch.

The formulation specifies the information structure, decision variables, accounting mechanisms, and performance objective of the sequential control problem. It does not impose a closed-form transition law for physical project progress and does not require the future transition distribution to be known in advance. Instead, next-period states are obtained from observed project-control data, empirical forecasting systems, simulation environments, or scenario-generation mechanisms. Therefore, the model should be interpreted as a sequential decision framework under observed state evolution. If a simulator, empirical transition model, or reinforcement-learning environment is supplied, the same formulation can be implemented using simulation-based optimization, approximate dynamic programming, or reinforcement learning.

4.12 Objective Function

The desired portfolio outcomes include:

- completing projects close to or before their planned finish dates;
- maximizing realized project and portfolio value;
- preserving expected profitability and avoiding uneconomic continuation;
- limiting external borrowing and financing costs;
- avoiding credit-limit violations and liquidity stress;
- reducing delay penalties and schedule exposure;
- avoiding unnecessary intervention expenditures;
- discouraging spending on projects with low expected continuation value.

These objectives may conflict. For example, aggressive acceleration can improve schedule performance but reduce profitability, while strict profitability control may lead to reduced funding for delayed projects. Therefore, the objective is expressed as a weighted multi-objective performance function whose weights reflect management preferences.

Let V_t denote the realized portfolio operating value in period t , before explicit intervention, financing, credit, and delay-penalty terms are applied. A clean accounting definition is

$$V_t = Y_t^P - \sum_{i \in P_t} e_{i,t}.$$

Equivalently,

$$V_t = \sum_{i \in P} Y_{i,t} - \sum_{i \in P_t} e_{i,t}.$$

This definition allows revenue from any project to be recognized in period t , including delayed receipts from completed projects, while expenditure is incurred only for projects that remain active. Intervention costs, financing costs, credit-violation costs, delay exposure, and terminal liquidated damages are kept as explicit penalty terms in the objective. This avoids double-counting and preserves accounting clarity.

The estimated continuation margin $\hat{\mu}_{i,t}$ summarizes the economic attractiveness of continuing project i at time t . For example, it may be defined as

$$\hat{\mu}_{i,t} = \frac{\widehat{CR}_{i,t}^{\text{rem}} - \widehat{C}_{i,t}^{\text{rem}} - \widehat{C}_{i,t}^{\text{delay}}}{\max\{\epsilon, \widehat{C}_{i,t}^{\text{rem}}\}},$$

where $\widehat{CR}_{i,t}^{\text{rem}}$ is expected remaining revenue, $\widehat{C}_{i,t}^{\text{rem}}$ is expected remaining execution cost, and $\widehat{C}_{i,t}^{\text{delay}}$ is expected delay-related cost, based on the latest project-control information.

Define the period-level portfolio reward as

$$R_t = \phi_V V_t - \phi_D D_t - \phi_{\text{act}} C_t^{\text{act}} - \phi_{\text{fin}} C_t^{\text{fin}} - \phi_{\text{Cr}} C_t^{\text{credit}} - \phi_{\text{low}} \sum_{i \in P_t} \frac{e_{i,t}}{\text{BAC}_i} \mathbf{1}\{\hat{\mu}_{i,t} < \underline{\mu}_i\}. \quad (4)$$

The sequential portfolio-control objective is then

$$\max_{\pi} \mathbb{E}_{\pi, \mathcal{M}} \left[\sum_{t=1}^H R_t - \phi_{\text{LD}} \sum_{i \in P} C_i^{\text{LD}} \right], \quad (5)$$

subject to the project-state observations, revenue timing, retention mechanism, credit mechanism, intervention mechanism, operational feasibility bounds, and information constraints defined above.

The expectation in (5) is taken over the trajectory distribution induced by the policy π and the underlying project-control environment $\mathcal{M}$. The formulation does not require $\mathcal{M}$ to be known analytically in advance. In a reinforcement-learning implementation, this expectation is approximated through sampled trajectories generated by interaction with a simulator, historical data environment, or calibrated stochastic project-control model.

In (4) and (5), V_t is realized portfolio operating value, D_t is aggregate period-level delay exposure, C_t^{act} is intervention cost, C_t^{fin} is financing cost, C_t^{credit} is credit-limit violation cost, and C_i^{LD} is the terminal liquidated damage cost for project i . The term $\hat{\mu}_{i,t}$ denotes the estimated continuation margin of project i , and $\underline{\mu}_i$ is the minimum acceptable continuation margin. The final period-level penalty discourages expenditure on projects whose expected continuation margin has fallen below the management-defined threshold.

The coefficients

$$\phi_V, \phi_D, \phi_{\text{act}}, \phi_{\text{fin}}, \phi_{\text{Cr}}, \phi_{\text{low}}, \phi_{\text{LD}} \geq 0$$

represent management preferences. Higher schedule-related weights represent a policy that prioritizes on-time delivery. Higher financing and credit weights represent a policy that avoids liquidity stress. Higher profitability and low-continuation-value weights represent a policy that prioritizes financial discipline and may reduce funding to projects with weak expected returns. The formulation is therefore not tied to a single managerial objective; rather, it provides a flexible structure that can be calibrated to the organizations risk appetite, contractual environment, and strategic priorities.

4.13 Assumptions, Scope, and Limitations

The formulation is based on the following assumptions:

- The project portfolio is fixed; project admission, bidding, and selection decisions are outside the scope.
- Contractual project attributes such as planned schedule, BAC, S-curve profile, milestone structure, and payment fractions are known inputs.
- The contractual S-curve is a baseline benchmark for planned expenditure and planned progress, not a deterministic law of actual project execution.
- Actual project progress is observed at each decision epoch and may deviate from the contractual plan.
- Earned value, SPI, CPI, remaining-duration estimates, intervention-persistence states, and progress observations are treated as project-control information available at each decision epoch.
- Budget allocation and intervention decisions affect future project performance through uncertain contractor response, but the realized effect is only observed at the next reporting epoch.
- Remaining-duration estimates, performance indices, and progress observations are treated as project-control information rather than outputs of a fixed deterministic production law.
- Planned project finish dates are fixed contractually, but actual project completion times are endogenous outcomes.
- A project remains active until realized physical progress reaches the completion threshold.
- Duration, cost, payment, contractor-performance, and intervention-effectiveness uncertainties are kept distribution-free at the formulation level.
- The formulation does not require known transition probabilities or known future uncertainty distributions in advance.
- Milestone payments are triggered by realized progress and recognized in discrete periods using a ceiling-based period mapping.

- Retention is withheld from interim progress payments and fully released with the final milestone at project completion.
- A separate defects liability period is not modeled because it lies beyond the active execution-focused planning horizon.
- Funding shortages are feasible through credit or external financing, but they generate financing costs and possible credit-limit penalties.
- Credit limits are soft constraints; violations are allowed but penalized through credit-cost functions.
- Corrective interventions may improve observed performance indicators, but their benefits are primarily captured through improved project outcomes rather than added as separate reward terms.
- The formulation defines the decision, accounting, and performance-evaluation structure. Implementation through simulation, stochastic optimization, approximate dynamic programming, or reinforcement learning requires an empirical, scenario-based, simulator-driven, or learned mechanism for generating next-period project states.

5 Methodology

5.1 Overview of the Proposed Solution Framework

5.2 Literature-Calibrated Synthetic Portfolio Generation

5.3 Environment Initialization

5.4 MDP Specification

5.5 State Representation

5.6 Action Space and Feasibility Handling

5.7 Transition Dynamics

5.8 Reward Function

5.9 PPO Training Procedure

5.10 Benchmark Policies and Evaluation Metrics

5.11 Forward Policy Rollout

6 Results and Analysis

6.1 Overview

[This section will be completed in future work after experimental implementation. Placeholder structure provided below.]

6.2 Solution Quality Comparison

[To be completed: Tables and figures showing:]

- Table: Expected NPV by method (RL, Greedy, MIP, SP, Perfect Info)

- Figure: NPV distribution boxplots
- Table: Budget utilization rates
- Figure: Risk-return tradeoff (NPV vs. variance)

Expected findings:

- RL outperforms deterministic MIP by $\sim 15\text{--}20\%$ in expected NPV
- RL matches or exceeds stochastic programming performance
- RL achieves $\sim 90\text{--}95\%$ of perfect-information upper bound

6.3 Computational Performance

[To be completed: Tables and figures showing:]

- Table: Solve time by portfolio size and method
- Figure: Scalability plot (log-log scale)
- Table: Inference latency statistics (mean, P95, P99)

Expected findings:

- RL inference: 100–200ms per decision (constant with N)
- MIP solve time: 15–45 minutes for $N = 10$, exponential growth
- Speedup factor: $10,000\times$ for $N = 10$, increasing with N

6.4 Sensitivity Analysis

[To be completed: Analysis of performance under varying conditions:]

Uncertainty Level:

- RL maintains robust performance across low/medium/high uncertainty
- Deterministic MIP degrades significantly under high uncertainty
- SP performance depends on scenario tree quality

Budget Constraint Tightness:

- RL learns effective prioritization under tight budgets
- Greedy heuristic fails to optimize across projects
- MIP optimal under loose budgets but slow to compute

Portfolio Composition:

- Performance consistent across 60/40 domestic/international mix
- Sensitivity to extreme compositions (80/20, 20/80) to be analyzed

6.5 Ablation Studies

[To be completed: Component-wise analysis:]

- Impact of rolling horizon length $H \in \{6, 12, 18, 24\}$
- Effect of reward function components (NPV, utilization, penalties)
- Importance of SPI-CPI correlation in training data
- Value of attention mechanism (if implemented)

6.6 Discussion

[To be completed: Interpretation of results, including:]

Why RL Outperforms Classical Methods:

- Learns robust policies from stochastic simulations
- Adapts to partial observability without explicit transition probabilities
- Generalizes across portfolio configurations

Computational Advantage:

- One-time training cost amortized over many deployments
- Inference time independent of portfolio size (within trained range)
- Enables real-time scenario analysis and Monte Carlo simulation

Limitations:

- Performance bounded by training data distribution
- Requires fine-tuning for company-specific patterns
- Black-box nature may limit practitioner trust (addressed by interpretability analysis)

7 Conclusion

7.1 Summary of Contributions

This paper establishes the foundational methodology for Reinforcement Learning-based project portfolio budgeting under cashflow uncertainty. Our work addresses a critical gap in operations research: the inability of classical optimization methods to handle sequential decision-making under partial observability in portfolio management.

We make four primary contributions:

(1) Literature-Calibrated Synthetic Data Generation: We developed a comprehensive framework for generating realistic project portfolios with empirically-grounded uncertainty distributions. All parameters are calibrated from published studies (318 megaprojects, multiple empirical analyses), ensuring scientific validity while addressing confidentiality constraints.

(2) Rolling Horizon Control Architecture: We proposed a rolling horizon framework that maintains fixed 12-period episodes while handling variable-duration portfolios

through sequential re-planning, preventing state-space explosion and enabling tractable RL training.

(3) Formal Problem Formulation: We provided the first rigorous mathematical formulation of portfolio budgeting as a POMDP with explicit state space, action space, transition dynamics, and reward function, all grounded in empirical project management literature.

(4) Complete Modeling Framework: We documented a modular portfolio modeling system covering composition, cashflows, uncertainty, and revenue plans, with all assumptions explicitly stated and future extensions identified.

7.2 Implications for Research and Practice

For Operations Research: This work demonstrates that RL can address problems where classical OR methods fail due to unknown transition probabilities and computational intractability. The rolling horizon architecture provides a template for applying RL to other long-horizon sequential decision problems.

For Project Management: The literature-calibrated synthetic data methodology enables reproducible research in domains with confidentiality constraints. Our framework can be extended to construction, infrastructure, R&D, and other project-intensive industries.

For Machine Learning: This application showcases RL’s potential in operations research domains, complementing recent successes in robotics, game playing, and control. The integration of domain knowledge (S-curves, milestone payments, SPI/CPI dynamics) with data-driven learning exemplifies hybrid AI approaches.

7.3 Limitations and Scope Boundaries

We deliberately adopted the *Simplest Model Principle* to establish a foundational framework. Key limitations include:

- **Aggregated cashflow modeling:** No resource-level detail (labor, materials, equipment)
- **Fixed portfolio composition:** 60/40 domestic/international mix assumed optimal
- **Milestone-based payments only:** Alternative structures (time-based, cost-plus) excluded
- **No project interdependencies:** Shared resources and technology spillovers not modeled
- **Synthetic data only:** Real-world validation pending company partnerships

Each limitation represents a natural extension for future work, documented with clear research directions.

7.4 Future Work

Immediate Next Steps (Current Research):

1. **RL Agent Implementation:** Select and train policy network (PPO, SAC, or Transformer-based)

2. **Baseline Comparison:** Implement MIP, stochastic programming, and greedy heuristics
3. **Experimental Evaluation:** Conduct comprehensive experiments across scenarios
4. **Sensitivity Analysis:** Analyze robustness to parameter variations

Medium-Term Extensions (6–12 Months):

1. **Fine-Tuning Protocol:** Develop methodology for adapting pretrained models to company-specific data
2. **Interpretability Analysis:** Apply attention mechanisms or SHAP values to explain allocation decisions
3. **Multi-Objective Optimization:** Extend to Pareto-optimal policies balancing NPV, risk, and strategic alignment
4. **Real-World Validation:** Partner with EPC contractors for pilot deployment

Long-Term Research Directions (1–3 Years):

1. **Resource-Level Modeling:** Disaggregate cashflows into labor, materials, equipment with detailed scheduling
2. **Dynamic Portfolio Selection:** Integrate project selection with execution optimization
3. **Multi-Agent RL:** Model interactions between contractor, client, and subcontractors
4. **Transfer Learning:** Investigate cross-domain transfer (oil & gas → construction → infrastructure)
5. **Hierarchical RL:** Develop two-level policies (strategic portfolio allocation + tactical project control)

7.5 Broader Impact

This research has potential impact beyond project portfolio management:

Methodological Contributions:

- Template for applying RL to OR problems with unknown transition dynamics
- Framework for literature-calibrated synthetic data generation
- Rolling horizon architecture for long-horizon sequential decision problems

Practical Applications:

- Real-time decision support for portfolio managers
- Scenario analysis for strategic planning
- Risk assessment through Monte Carlo simulation
- Training tool for project management education

Open Science:

- Complete codebase and datasets released as open source
- Reproducible research enabling community validation and extension
- Benchmark problem for RL algorithm development

7.6 Concluding Remarks

Project portfolio budgeting under uncertainty represents a critical challenge for capital-intensive industries. Traditional optimization methods fail due to computational intractability and unknown state transition probabilities. This paper establishes the foundational methodology for addressing this challenge through Reinforcement Learning with rolling horizon control.

By developing a literature-calibrated synthetic data generation framework, we overcome the confidentiality barrier that has prevented progress in this domain. Our formal problem formulation and complete modeling framework provide a rigorous foundation for future research.

The path forward is clear: implement and evaluate RL agents, compare against classical baselines, and develop fine-tuning protocols for company-specific deployment. We invite the research community to build upon this foundation, extending the methodology to related domains and advancing the state of the art in RL-based operations research.

The complete modeling framework, instance generator, and documentation will be released as open-source software upon publication, enabling reproducible research and practical deployment in this emerging field.

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References

- Archer, N. P. and Ghasemzadeh, F. (1999). An integrated framework for project portfolio selection. *International Journal of Project Management*, 17(4):207–216.
- Ballestín, F. and Blanco, R. (2015). Resource-constrained multi-project scheduling problem: A survey. *European Journal of Operational Research*, 249(2):412–434.
- Barraza, G. A., Back, W. E., and Mata, F. (2007). Probabilistic forecasting of project performance using stochastic S curves. *Journal of Construction Engineering and Management*, 133(1):49–59.
- Bengio, Y., Lodi, A., and Prouvost, A. (2021). Machine learning for combinatorial optimization: A methodological tour d’horizon. *European Journal of Operational Research*, 290(2):405–421.
- Chand, S., Hsu, V. N., and Sethi, S. (2002). Rolling horizon procedures for dynamic parallel machine scheduling with sequence-dependent setup times. *International Journal of Production Research*, 40(18):4855–4879.

- Chen, Z., Li, H., and Wong, C. T. W. (2018). Reinforcement learning for construction project scheduling under resource uncertainty. *Automation in Construction*, 93:127–141.
- Cioffi, D. F. (2005). A tool for managing projects: An analytic parameterization of the S-curve. *International Journal of Project Management*, 23(3):215–222.
- Cooper, R. G., Edgett, S. J., and Kleinschmidt, E. J. (2001). *Portfolio Management for New Products*. Basic Books.
- Cui, Q., Hastak, M., and Halpin, D. (2010). A systems approach to project cash flow management. *Journal of Construction Engineering and Management*, 136(12):1281–1290.
- Doerner, K. F., Gutjahr, W. J., Hartl, R. F., Strauss, C., and Stummer, C. (2006). Pareto ant colony optimization with ILP preprocessing in multiobjective project portfolio selection. *European Journal of Operational Research*, 171(3):830–841.
- Elshaer, R. (2014). A genetic algorithm for multi-objective project portfolio selection. *International Journal of Project Management*, 32(4):546–558.
- Fleming, Q. W. and Koppelman, J. M. (2016). *Earned Value Project Management*. Project Management Institute.
- Flyvbjerg, B., Holm, M. K. S., and Buhl, S. L. (2003). How common and how large are cost overruns in transport infrastructure projects? *Transport Reviews*, 23(1):71–88.
- Flyvbjerg, B., Holm, M. S., and Buhl, S. (2002). Underestimating costs in public works projects: Error or lie? *Journal of the American Planning Association*, 68(3):279–295.
- Golenko-Ginzburg, D. and Gonik, A. (1997). Stochastic network project scheduling with non-consumable limited resources. *International Journal of Production Economics*, 48(1):29–37.
- Gutjahr, W. J. and Froeschl, K. A. (2013). Project portfolio selection under uncertainty with outsourcing opportunities. *Flexible Services and Manufacturing Journal*, 25:255–281.
- Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. (2018). Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In *Proceedings of the 35th International Conference on Machine Learning*, pages 1861–1870.
- Hubbs, C. D., Li, C., Sahinidis, N. V., Grossmann, I. E., and Wassick, J. M. (2020). Deep reinforcement learning for industrial process control: A case study in chemical production. *Computers & Chemical Engineering*, 141:106982.
- Jordon, J., Szpruch, L., Houssiau, F., Bottarelli, M., Cherubin, G., Maple, C., Cohen, S. N., and Weller, A. (2022). Synthetic Data—What, Why and How?
- Kaka, A. P. and Price, A. D. F. (1993). Towards more flexible and accurate cash flow forecasting. *Construction Management and Economics*, 11(1):35–44.
- Kenley, R. and Wilson, O. D. (1986). A construction project cash flow model—an idiographic approach. *Construction Management and Economics*, 4(3):213–232.

- Khanzadi, M., Nasirzadeh, F., and Alipour, M. (2018). Integrating project portfolio selection and scheduling under uncertainty. *Journal of Construction Engineering and Management*, 144(2):04017106.
- Killen, C. P., Hunt, R. A., and Kleinschmidt, E. J. (2008). Project portfolio management for product innovation. *International Journal of Quality & Reliability Management*, 25(1):24–38.
- Klerides, E. and Hadjiconstantinou, E. (2019). A stochastic programming approach for project portfolio selection under uncertainty. *Annals of Operations Research*, 274(1):361–382.
- Kolisch, R. and Sprecher, A. (1997). PSPLIB—a project scheduling problem library: OR software—ORSEP operations research software exchange program. *European Journal of Operational Research*, 96(1):205–216.
- Lambrechts, O., Demeulemeester, E., and Herroelen, W. (2008). Proactive and reactive strategies for resource-constrained project scheduling with uncertain resource availabilities. *Journal of Scheduling*, 11(2):121–136.
- Liesiö, J., Mild, P., and Salo, A. (2007). Portfolio decision analysis for robust project selection. *European Journal of Operational Research*, 181(3):1488–1505.
- Lipke, W., Zwikael, O., Henderson, K., and Anbari, F. (2009). Prediction of project outcome: The application of statistical methods to earned value management and earned schedule performance indexes. *International Journal of Project Management*, 27(4):400–407.
- Love, P. E. D., Sing, C.-P., Wang, X., Edwards, D. J., and Odeyinka, H. (2013). Probability distribution fitting of schedule overruns in construction projects. *Journal of the Operational Research Society*, 64(8):1231–1247.
- Love, P. E. D., Sing, M. C.-P., Ika, L. A., and Newton, S. (2016). The relationship between schedule and cost performance in construction projects. *Journal of Construction Engineering and Management*, 142(9):04016038.
- Martinsuo, M. (2013). Project portfolio management in practice and in context. *International Journal of Project Management*, 31(6):794–803.
- Morrow, E. W. (2011). *Industrial Megaprojects: Concepts, Strategies, and Practices for Success*. Wiley.
- Puterman, M. L. (1994). *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons.
- Sahin, F., Narayanan, A., and Robinson, E. P. (2013). Rolling horizon planning in supply chains: Review, implications and directions for future research. *International Journal of Production Research*, 51(18):5413–5436.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal Policy Optimization Algorithms.
- Sethi, S. P. and Thompson, G. L. (1991). A survey of the maximum principles for optimal control problems with state constraints. *SIAM Review*, 33(2):181–218.

- Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press.
- Vanhoucke, M. (2012). An integrated project management and control framework. *Journal of Modern Project Management*, 1(1):14–27.
- Wauters, T., Kinable, J., Smet, P., Vancroonenburg, W., Vanden Berghe, G., and Verstichel, J. (2016). Learning agents for the multi-mode project scheduling problem. *Journal of the Operational Research Society*, 67(8):1031–1046.
- Xu, L., Skoularidou, M., Cuesta-Infante, A., and Veeramachaneni, K. (2019). Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32.

A Mathematical Proofs

[Placeholder for mathematical proofs. To be completed if needed.]

A.1 Proof of Rolling Horizon Optimality Conditions

[To be added: Conditions under which rolling horizon policy approximates infinite-horizon optimal policy]

A.2 Proof of State Space Completeness

[To be added: Demonstration that state representation captures all decision-relevant information]

B Parameter Calibration Details

B.1 Complete Parameter Tables

This appendix provides comprehensive parameter tables for all model components, with literature citations and calibration details.

B.1.1 Portfolio Composition Parameters

Table 2: Portfolio Composition Model Parameters

Parameter	Value	Unit	Source
Domestic weight	0.60	fraction	Khanzadi et al. (2018)
International weight	0.40	fraction	Khanzadi et al. (2018)
Portfolio size min	8	projects	Merrow (2011), CII (2019)
Portfolio size max	18	projects	Merrow (2011), CII (2019)

Table 3: Profit Margin Distribution Parameters

Parameter	Domestic	International	Source
Mean μ	0.10	0.14	Khanzadi et al. (2018)
Std Dev σ	0.012	0.030	Khanzadi et al. (2018)
Lower bound a	0.08	0.12	Iran Budget Law
Upper bound b	0.12	0.16	Market analysis

Table 4: Project BAC Lognormal Distribution

Parameter	Value	Source
$\mu_{\log}$	5.01	Merrow (2011)
$\sigma_{\log}$	0.69	Merrow (2011)
Median BAC	\$150M	Merrow (2011)
Mean BAC	\$280M	Merrow (2011)
Min BAC	\$50M	Industry practice
Max BAC	\$5B	Industry practice

B.1.2 Profit Margin Parameters**B.1.3 Project BAC Parameters****B.1.4 S-Curve Parameters****B.1.5 Performance Uncertainty Parameters****B.1.6 Duration Model Parameters****B.1.7 Milestone Payment Structure****B.2 Validation Results**

[To be completed: Statistical validation of generated instances against target distributions]

B.2.1 Distributional Validation

[Tables showing sample statistics vs. target values for all key variables]

B.2.2 Correlation Validation

[Correlation matrices comparing generated data to empirical values]

B.2.3 Portfolio-Level Validation

[Aggregate statistics for generated portfolios]

C Instance Generator Algorithm**C.1 Instance Generator Algorithm**

This appendix provides the complete pseudocode for the portfolio instance generator.

Table 5: S-Curve Beta Distribution Base Parameters

Parameter	Value	Source
α_{base}	2.5	Cioffi (2005)
β_{base}	2.0	Cioffi (2005)
Peak time τ^*	0.60	Kenley & Wilson (1986)
Size exponent	0.15	Barraza et al. (2007)
Duration exponent	0.12	Barraza et al. (2007)
Residual noise σ	0.10	Calibrated

Table 6: SPI and CPI Distribution Parameters

Parameter	SPI	CPI	Correlation	Source
α	2.8	3.2	–	Morrow (2011)
β	2.2	1.8	–	Love et al. (2016)
Lower bound	0.3	0.5	–	Empirical range
Upper bound	1.3	1.2	–	Empirical range
Mean	0.86	0.87	–	Calibrated
Std Dev	0.18	0.14	–	Calibrated
$\rho(\text{SPI, CPI})$	–	–	0.68	Love et al. (2016)

Algorithm 1 Portfolio Instance Generator**Require:** Portfolio size range $[N_{\min}, N_{\max}]$, parameter tables**Ensure:** Portfolio instance $\mathcal{P}$ with all attributes

```

1:
2: Stage 1: Portfolio Composition
3:  $N \leftarrow \text{UniformInt}(N_{\min}, N_{\max})$ 
4:  $N_{\text{domestic}} \leftarrow \lfloor 0.6 \times N \rfloor$ 
5:  $N_{\text{international}} \leftarrow N - N_{\text{domestic}}$ 
6:
7: Stage 2: Project Characteristics
8: for  $i = 1$  to  $N$  do
9:   Assign category: domestic or international
10:  Sample  $\log(\text{BAC}_i) \sim \mathcal{N}(\mu_{\log}, \sigma_{\log})$ 
11:  Clip  $\text{BAC}_i$  to  $[\$50M, \$5B]$ 
12:  Sample  $\pi_i \sim \text{TruncatedNormal}(\mu_{\pi}, \sigma_{\pi}, a_{\pi}, b_{\pi})$ 
13:  Compute  $R_i \leftarrow \text{BAC}_i \times (1 + \pi_i)$ 
14:  Sample  $D_i \leftarrow D_{\text{base}} \times (\text{BAC}_i / \text{BAC}_{\text{ref}})^{0.35} \times \epsilon_D$ 
15:  Clip  $D_i$  to  $[12, 60]$  months
16: end for
17:
18: Stage 3: S-Curve Parameters
19: for  $i = 1$  to  $N$  do
20:   Compute size multiplier:  $m_{\text{size}} \leftarrow (\text{BAC}_i / \text{BAC}_{\text{median}})^{0.15}$ 
21:   Compute duration multiplier:  $m_{\text{dur}} \leftarrow (D_i / D_{\text{median}})^{0.12}$ 
22:   Get category multiplier  $m_{\text{cat}}$  (0.9 for domestic, 1.1 for international)
23:   Sample risk level: low or high
24:   Get risk multipliers  $m_{\text{risk}}^{\alpha}, m_{\text{risk}}^{\beta}$ 
25:   Sample  $\epsilon_{\alpha}, \epsilon_{\beta} \sim \text{LogNormal}(0, 0.10)$ 
26:    $\alpha_i \leftarrow \alpha_{\text{base}} \times m_{\text{size}} \times m_{\text{cat}} \times m_{\text{risk}}^{\alpha} \times \epsilon_{\alpha}$ 
27:    $\beta_i \leftarrow \beta_{\text{base}} \times m_{\text{dur}} \times m_{\text{cat}} \times m_{\text{risk}}^{\beta} \times \epsilon_{\beta}$ 
28:   Clip  $\alpha_i$  to  $[1.2, 5.0]$ ,  $\beta_i$  to  $[1.2, 5.5]$ 
29: end for
30:
31: Stage 4: Cashflow Profiles
32: for  $i = 1$  to  $N$  do

```

Table 7: Project Duration Model Parameters

Parameter	Value	Source
D_{base}	24 months	Merrow (2011)
BAC_{ref}	\$150M	Merrow (2011)
Scale exponent	0.35	Merrow (2011)
Noise σ_D	0.15	Calibrated
Min duration	12 months	Industry practice
Max duration	60 months	Industry practice

Table 8: Standard Milestone Payment Weights

Milestone	Progress Threshold	Payment Weight
Advance	0%	10%
Milestone 1	25%	20%
Milestone 2	50%	30%
Milestone 3	75%	30%
Final	100%	10%

C.2 Validation Algorithm

Algorithm 2 Instance Validation

Require: Generated portfolio instance $\mathcal{P}$, target statistics

Ensure: Validation report (pass/fail for each check)

```

1:
2: Level 1: Distributional Validation
3: for each variable  $X$  in  $\{\text{BAC}, \pi, D, \alpha, \beta, \text{SPI}, \text{CPI}\}$  do
4:   Compute sample mean  $\bar{X}$ , std dev  $s_X$ 
5:   Check:  $|\bar{X} - \mu_{\text{target}}|/\mu_{\text{target}} < 0.05$  ▷ 5% tolerance
6:   Check:  $|s_X - \sigma_{\text{target}}|/\sigma_{\text{target}} < 0.10$  ▷ 10% tolerance
7: end for
8:
9: Level 2: Correlation Validation
10: Compute  $\rho(\text{SPI}, \text{CPI})$ 
11: Check:  $|\rho - 0.68| < 0.05$ 
12:
13: Level 3: Portfolio-Level Validation
14: Compute total portfolio BAC:  $\sum_{i=1}^N \text{BAC}_i$ 
15: Compute domestic fraction:  $\sum_{i \in \text{domestic}} \text{BAC}_i / \sum_{i=1}^N \text{BAC}_i$ 
16: Check: domestic fraction  $\in [0.55, 0.65]$  ▷ 60%  $\pm$  5%
17:
18: Level 4: Sanity Checks
19: for  $i = 1$  to  $N$  do
20:   Check:  $\text{BAC}_i > 0, D_i > 0, \pi_i > 0$ 
21:   Check:  $\alpha_i \in [1.2, 5.0], \beta_i \in [1.2, 5.5]$ 
22:   Check:  $\text{SPI}_i \in [0.3, 1.3], \text{CPI}_i \in [0.5, 1.2]$ 
23:   Check:  $\sum_m w_m = 1.0$  ▷ Milestone weights sum to 100%
24: end for
25:
26: return Validation report

```

C.3 Implementation Notes

Computational Complexity: The instance generator has time complexity $O(N \times D_{\max})$ where N is portfolio size and $D_{\max}$ is maximum project duration. For typical values ($N = 15, D_{\max} = 60$), generation takes < 1 second per instance.

Reproducibility: All random number generation uses seeded PRNGs (NumPy default_rng with explicit seeds). Seeds are logged for each generated instance.

Parallelization: Instance generation is embarrassingly parallel. For large datasets (10,000+ instances), use multiprocessing with one worker per CPU core.

Output Format: Instances are serialized as JSON or HDF5 files with standardized schema for interoperability.